

Multiple Choice (10 marks)

1. 1. _____ framework made cracking of vulnerabilities easy like point and click.
 - (a) .Net
 - (b) Metasploit
 - (c) Zeus
 - (d) Ettercap
2. Old operating systems like _____ and NT-based systems have buffer-overflow attack a common vulnerability.
 - (a) Windows 7
 - (b) Chrome
 - (c) IOS 12
 - (d) UNIX
3. Which of the following is not an example of presentation layer issues?
 - (a) Poor handling of unexpected input can lead to the execution of arbitrary instructions
 - (b) Unintentional or ill-directed use of superficially supplied input
 - (c) Cryptographic flaws in the system may get exploited to evade privacy
 - (d) Weak or non-existent authentication mechanisms
4. Buffer-overflow may remain as a bug in apps if _____ are not done fully.
 - (a) boundary hacks
 - (b) memory checks
 - (c) boundary checks
 - (d) buffer checks
5. A _____ may be a hidden part of a program, a separate infected program a Trojan in disguise of an executable or code in the firmware of any system's hardware.
 - (a) crypter
 - (b) virus
 - (c) backdoor
 - (d) key-logger

True / False (10 marks)

Answer True or False

6. True or False: Escaping queries is an essential technique to prevent SQL injection vulnerabilities in web applications.
7. True or False: A session symmetric key can be reused multiple times for different sessions between the same parties.
8. True or False: Message confidentiality ensures that only the intended sender and receiver can access the content of the communication.
9. True or False: WPA 2 is weaker than WEP, making it less secure for wireless communications.
10. True or False: The dark web is a segment of the Deep Web that is intentionally hidden and inaccessible through standard web browsers.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the SYN stealth scan in Nmap, analyzing its advantages and limitations compared to other TCP scanning techniques in terms of stealth and effectiveness in cybersecurity.
12. Discuss the significance and process of the 4-way handshake in wireless authentication, analyzing its role in ensuring secure communication between a user and an access point.

End of Paper